30/1/2025

CSE704

Automata & logic

FSA $\cong$ MSO

(equivalence)

MSO over Σ: $x, y, \ldots \; ; X, Y, \ldots$

Syntax: $Q_a(x)$, $x < y$, $x \in X$

$\neg, \vee, \wedge$

$\exists x, \forall x, \exists X, \forall X$

Semantics: Formulas are interpreted over words in Σ^*.

Each position in a word is assigned to a variable by the underlying intepretation

Each set of positions is assigned to a set variable.

First-order logic over numbers:

A general first order logic has

constants
variables
functions
relations
logical operators
quantifiers

General first order logic is undecidable. i.e., given a formula φ

the problem of checking if φ is <u>satisfiable</u> is undecidable.

We restrict ourselves to

$$FO(\mathbb{N}, <, +) \qquad FO(\mathbb{N}, <, +, -)$$

Function

$-$ already present in the coefficients

Domain is the set of natural numbers $\{0, 1, 2, \ldots\}$

Relation ($>, \leq, \geq, =, \neq$ are all there due to the presence of logical operators).

Sample formulas:

$$x+y$$

$$2x+5y$$

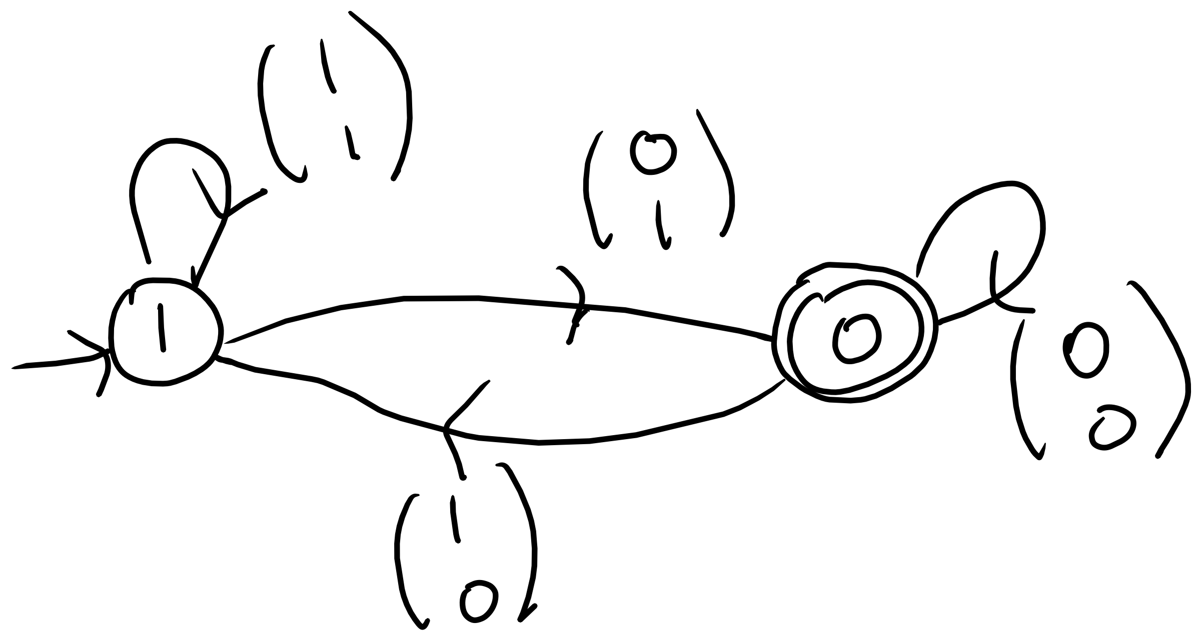
$$x-3y$$

$$3x-4y+z<3$$

x^2+2 — Not expressible in our logic(s)

$3x^3-4y^2+z$ - Again not expressible.

Exercise: Solve $y = 2x+1$ using Presburger's automaton construction.



x, y

$$\begin{pmatrix}0\\0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}$$
$$\begin{pmatrix}1\\1\end{pmatrix}$$

$$\begin{array}{l|cc} x = 1 & 0 & 1 \\ y = 3 & 1 & 1 \end{array}$$

$$\uparrow \quad \uparrow$$

$x = 7$
$y = 15$
$$\begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}$$

Exercise: Solve $x + y + z = 7$

Solve $2x - y + 3z = 5$